Name of Student: Shreyasi Mishra	**PRN: 25070122294**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

```
class Student {

    int rollNo;

    String name;

    double marks;

    // Default Constructor

    Student() {

        rollNo = 0;

        name = "Unknown";

        marks = 0.0;

    }

    // Parameterized Constructor

    Student(int rollNo, String name, double marks) {
```

```
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }
    // Copy Constructor
    Student(Student s) {
        this.rollNo = s.rollNo;
        this.name = s.name;
        this.marks = s.marks;
    }
    // Method to display student details
    void display() {
        System.out.println("Roll No : " + rollNo);
        System.out.println("Name    : " + name);
        System.out.println("Marks   : " + marks);
        System.out.println();
    }
    public static void main(String[] args) {
        // Using Default Constructor
        Student s1 = new Student();
        // Using Parameterized Constructor
        Student s2 = new Student(101, "Shreyasi", 92.5);
        // Using Copy Constructor
        Student s3 = new Student(s2);
        System.out.println("Default Constructor:");
        s1.display();
        System.out.println("Parameterized Constructor:");
```

```
        s2.display();
        System.out.println("Copy Constructor:");
        s3.display();
    }
}
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```
public class MobileInventory {
    int id;
    String brand;
    String model;
    double price;
    // Default Constructor
    MobileInventory() {
        id = 0;
        brand = "Unknown";
        model = "Unknown";
        price = 0.0;
    }
    // Parameterized Constructor
    MobileInventory(int id, String brand, String model, double price) {
        this.id = id;
        this.brand = brand;
        this.model = model;
        this.price = price;
    }
    // Copy Constructor
```

```
    MobileInventory(MobileInventory m) {
        this.id = m.id;
        this.brand = m.brand;
        this.model = m.model;
        this.price = m.price;
    }
    // Display Method
    void display() {
        System.out.println("Mobile ID : " + id);
        System.out.println("Brand     : " + brand);
        System.out.println("Model     : " + model);
        System.out.println("Price     : Rs. " + price);
        System.out.println();
    }
    public static void main(String[] args) {
        MobileInventory m1 = new MobileInventory();
            MobileInventory m2 = new MobileInventory(101, "Samsung", "Galaxy
S24", 74999);
        MobileInventory m3 = new MobileInventory(m2);
        System.out.println("Default Constructor:");
        m1.display();
        System.out.println("Parameterized Constructor:");
        m2.display();
        System.out.println("Copy Constructor (Duplicate Record):");
        m3.display();
    }
}
```

Output

Output

```
Default Constructor:
Roll No : 0
Name    : Unknown
Marks   : 0.0

Parameterized Constructor:
Roll No : 101
Name    : Shreyasi
Marks   : 92.5

Copy Constructor:
Roll No : 101
Name    : Shreyasi
Marks   : 92.5
```

Output

```
Default Constructor:
Mobile ID : 0
Brand     : Unknown
Model     : Unknown
Price     : Rs. 0.0

Parameterized Constructor:
Mobile ID : 101
Brand     : Samsung
Model     : Galaxy S24
Price     : Rs. 74999.0

Copy Constructor (Duplicate Record):
Mobile ID : 101
Brand     : Samsung
Model     : Galaxy S24
Price     : Rs. 74999.0
```

Conclusion

Through this assignment, we understood the concept of constructors in Java and demonstrated how default constructors initialize objects with predefined values, parameterized constructors allow custom initialization with user-defined arguments, and copy constructors create a new object as an exact duplicate of an existing one. Each type of constructor serves a distinct purpose in object initialization, collectively providing flexibility and control over how objects are created in an object-oriented program. This assignment reinforced the importance of constructors as a fundamental mechanism in Java for ensuring objects are properly initialized before use.